

Problem 07

Let $Q = \{\pm 1, \pm i, \pm j, \pm k\}$ be the quaternion group, with $i^2 = j^2 = k^2 = ijk = -1$. Construct a surjective homomorphism

$$\phi : Q \longrightarrow \mathbb{Z}_2 \times \mathbb{Z}_2,$$

where the target group uses componentwise addition modulo 2. Give the image of each element of Q , and determine $\ker \phi$ and $\text{im } \phi$.

Problem 08

For $N \geq 2$, let $\bar{r}$ denote the residue class of r in $\mathbb{Z}_N$, and set

$$\mu_N = \{z \in \mathbb{C} : z^N = 1\}, \quad \psi(\bar{r}) = e^{2\pi i r/N}.$$

For $k \in \mathbb{Z}$, define $m_k(\bar{r}) = \overline{kr}$ and $P_k(z) = z^k$.

(1) Show that m_k and P_k are well-defined homomorphisms, and that ψ is an isomorphism from $\mathbb{Z}_N$ to μ_N .

(2) Show that the diagram

$$\begin{array}{ccc} \mathbb{Z}_N & \xrightarrow{m_{k_1}} & \mathbb{Z}_N \\ \psi \downarrow & & \downarrow \psi \\ \mu_N & \xrightarrow{P_{k_2}} & \mu_N \end{array}$$

commutes if and only if $k_1 \equiv k_2 \pmod{N}$.

Problem 09

Let $SO(3)$ act on the set $X = S^2 \times S^2$ of ordered pairs of unit vectors in $\mathbb{R}^3$ by simultaneous rotation,

$$R \cdot (\vec{u}, \vec{v}) = (R\vec{u}, R\vec{v}).$$

- (1) Show that two ordered pairs $(\vec{u}, \vec{v})$ and $(\vec{u}', \vec{v}')$ lie in the same orbit if and only if $\vec{u} \cdot \vec{v} = \vec{u}' \cdot \vec{v}'$.
- (2) Find the stabilizer of $(\vec{u}, \vec{v})$ when $\vec{u}$ and $\vec{v}$ are parallel and when they are not parallel. Which orbits are free?
- (3) For $\vec{u} \cdot \vec{v} = 0$, construct the unique rotation R satisfying $R\hat{e}_1 = \vec{u}$ and $R\hat{e}_2 = \vec{v}$. Show that $R \mapsto (R\hat{e}_1, R\hat{e}_2)$ identifies $SO(3)$ with the orbit of perpendicular pairs, as predicted by orbit-stabilizer.

Problem 10

Let G act linearly on complex vector spaces V and W . Let $\text{Hom}_{\mathbb{C}}(V, W)$ be the vector space of complex-linear maps from V to W . For $T \in \text{Hom}_{\mathbb{C}}(V, W)$ define

$$(g \cdot T)(v) = g \cdot (T(g^{-1} \cdot v)).$$

- (1) Verify that this defines a linear action of G on $\text{Hom}_{\mathbb{C}}(V, W)$.
- (2) Show that T is fixed by every $g \in G$ if and only if it is an equivariant map, meaning $T(g \cdot v) = g \cdot T(v)$ for all g, v .
- (3) For the usual action of $SU(2)$ on $V = W = \mathbb{C}^2$, show that every complex-linear equivariant map has the form $T(\vec{z}) = \alpha \vec{z}$ for some $\alpha \in \mathbb{C}$. You may use the matrices

$$\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \quad (\theta \in \mathbb{R}), \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Problem 11

Let D be the subgroup of diagonal matrices in $SU(2)$,

$$D = \left\{ \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} : |z| = 1 \right\}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

- (1) Find the centralizer $C_{SU(2)}(D)$.
- (2) Show that the normalizer is $N_{SU(2)}(D) = D \cup JD$.
- (3) Determine how each of the two sets D and JD acts on D by conjugation. Show that every element of JD exchanges the two diagonal entries and has order 4.

Problem 12

Let a finite group G act transitively on a finite set X with $|X| > 1$. For $g \in G$, write $\text{Fix}_X(g) = \{x \in X : g \cdot x = x\}$.

- (1) Count the pairs $(g, x) \in G \times X$ satisfying $g \cdot x = x$, first by fixing g and then by fixing x . Use orbit-stabilizer to prove

$$\sum_{g \in G} |\text{Fix}_X(g)| = |G|.$$

- (2) Deduce that some $g \in G$ has no fixed points in X .